NEURAL NETWORKS AND EVOLUTIONARY ALGORITHMS IN MUSIC COMPOSITION

ABSTRACT

In this paper we describe a system developed to generate short musical phrases in the same style as a set of training melodies. The system uses an ensemble of neural networks to rate the similarity of a generated musical phrase to a set of human composed phrases. Given this rating a genetic algorithm is used to effectively 'search' for other similar phrases. Preliminary evaluations indicate that the proposed approach shows promise. A limitation of the system is the relatively simple representation of musical phrases employed and the processing time required to use this technique on longer phrases.

1. INTRODUCTION

There has been much research into automatic computer music composition, and several techniques and approaches proposed, including the application of musical theories [1], deconstruction and intelligent reconstruction of compositions [2], transition tables [3] and cellular automaton [4] for example. Neural networks and genetic algorithms are also widely used and the application of these techniques is the focus of this paper.

The program GenJam [5] improvises jazz music using genetic algorithms using a human judge as it's fitness function. For the representation scheme music solos are broken into phrases which are then made up of four measures. A human judge listens to the generated solo and presses one or more 'g's for good or, one or more'b's for bad. From these human judgements the genetic algorithm selects measures and phrases to survive and reproduce. The phrases are then modified by musically meaningful rules which aim to create better offspring. Such modification rules for measures include transposition, inversion and reversion. For phrases modification rules include reversal, rotation and "lick thinner" which substitutes a random phrase in the measure for the most common phrase in the entire phrase population. After many generation of selection, reproduction and modification, solos are output.

Neurogen [6] also uses a genetic algorithm, however it divides its music compositions into building blocks of rhythm, melody and harmony to be judged by an independent neural networks rather than a human. Those musical phrases selected by the neural network may reproduce, but rather than the musical phrase being modified by rules informed by existing music theories a random modification rule is used.

Other projects have used a different approach in which the system attempts to complete a given musical phrase by predicting the next notes using a neural network. This prediction is the same concept as used for transition tables by which the use of neural network is an extension of [7]. The project "Learning the Long-Term Structure of the Blues" [8] (LTSM) uses this idea: it learns to predict the next notes from the previous notes using a neural network. Then given a seed of a few notes it then composes the rest of a piece using its predictions. This step by step composition might seem incapable of capturing capture long term features within music. In order to resolve this a Long Short-Term Memory [9] (LSTM) neural network architecture which is long term memory of past input allowing it to take in to consideration long term structure.

The project CONCERT [10] has a similar approach to LTSM as it predicts the next note given the previous using a neural network. However it does not use LSTM to help capture long term structure but rather a reduced description of the compositions. The individual pitches are "smoothed and compressed" so that events at a course time scale are more explicit. As the representation is also more compact it allows for a longer timescale to be considered at a time helping again in learning long term structure.

GenJam using human guided genetic algorithms with LTSM and CONCERT using predicting neural networks represent two common approaches. The less common approach of Neurogen using neural network guided genetic algorithms is most similar to the project we describe here.

1.1. Aim

The aim of this project is to develop a simple proof-of-concept system using neural networks in a novel way, that can be extended if desired. The aim of the system described in this paper is to compose single-voiced short phrases. These phrases should sound similar, to human ears, to provided training compositions. This includes emulating structure or context sensitive rules, if any, from the provided training compositions. For example if the provided compositions

have rising arpeggios or call and response, then those should be present in generated compositions. The system should learn how to generate similar compositions, removing the need for an expert. The system should also be able to generate its phrases with minimal assumptions on the nature of music so that it will be flexible enough to be used in a wide range of musical contexts. In other words, given examples of a new style of composition the system will be able to learn and generate similar compositions.

Initially, we have limited our system to single voice phrases to avoid the complexity of multiple voices and longer composition. The task of emulating any structure has been explicitly noted as this is a common shortcoming of computer generated music. In the case of predictive neural network projects this shortcoming is often due to the neural network looking at a small number of prior notes or events in order to decide what should come next. Notes further back are ignored and hence global structure is lacking. This is somewhat remedied by using LSTM recurrent neural networks which have a degree of memory such as the LTSM project, or by a reduced description representation scheme as CONCERT.

This paper's project hopes to avoid the problem completely by having all notes within a composition, rather that a few local notes, as input to a neural network. In this way the context considered by the neural network is of the whole piece. This is a similar approach as NeuroGen. Whilst Neurogen treats phrases as building blocks of rhythm, melody and harmony, this project deals with a melody as a whole. Neurogen deals with a smaller palette in terms of pitch. Where Neurogen is constrained to eight consecutive notes on the C major scale, this project is constrained to twenty five notes on the chromatic scale. The most significant change is the method of evaluation of musical phrases where this project introduces an iterative process which builds a growing ensemble of neural networks.

2. OVERVIEW

The system will be emulating a set of single voice human compositions, which will be used for training the system. These compositions are made up of successive notes of particular pitch and duration. The compositions are split into phrases of sixteen beats to be emulated, or four bars of common time. The phrases are then split again into slices of quaver length. Each slice represents a portion of time, whose value represents either a new note's pitch, a hold of the previous note or a rest. Pitches themselves are represented using a numerical value where middle C is 0, C sharp 1, D 2 and so on.

To generate phrases a genetic algorithm is used. A population of random phrases is first created. The phrases are then judged to be more or less similar to the given compositions. Those more similar are deemed to be "fitter". Within the population the fittest phrase can then reproduce with crossover and mutations forming the next generation, while "unfit" phrases do not. Each successive generation then should have "fitter", of more similar, musical phrases.

In order to judge the fittest phrases a neural network is used. The neural network is trained using the prepared compositions along with generated random phrases. The random generated phrases provide a counter example and consists of randomly selected note values. The neural network's task is then to give the probability of a given phrase of belonging to the class of training compositions. This probability can be each phrase's fitness rating.

However this neural network generation might not lead an accurate judge of musical phrases. This is as a neural network trained with random and training phrases may differentiate between each class by limited or even non-musical features. For example it might be possible to differentiate between random and human phrases simply by the frequency of a note, the mean values of all notes or mode of the starting interval. While these could be features of the human phrases they do not encompass enough features to direct generation of musical phrases. In turn such a judge used as fitness function in a genetic algorithm is unlikely to produce human phrase similar compositions, as was found when tried.

To overcome this issue an iterative process is used, illustrated in figure 1. At the first iteration, a single neural network is trained with the random and human phrases. The genetic algorithm then generates new phrases. For the second iteration two neural networks are used, each with its own purpose.

The first neural network's aim is to find features that have not yet been found, features of human phrases not found in the generated and random phrases. To do this neural network is trained with the human phrases as one class and, generated and random phrases as the counter-class. The second neural network aim is to find features already found, shared features of human and generated phrases not found in the random phrases. To do this the neural network is trained with human and generated phrases as one class and random phrases as the counter-class.

This results of the neural network are again used to guide the genetic algorithm, and generate the next new set of generated phrases.

The following iterations continue to use an ensemble of neural networks to find features found in each generated set and the final neural network to find features not yet found. Features found within each generated phrases are learnt by using human phrases, all later generated sets and the generated set itself as examples, and all other phrases, being earlier generated sets and the random sets as counter-examples. Features yet to be found are learnt by using human phrases as an example and all other phrases as counter-examples. In this way the ensemble can remember what features it has

learnt before while looking for new characteristics.

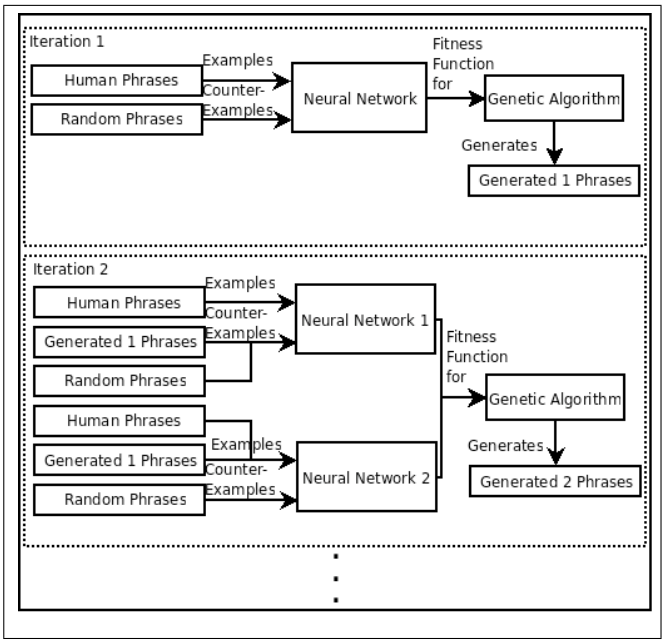


Figure 1. The iterative process as described, with each xth iteration training x neural networks.

3. IMPLEMENTATION DETAILS

3.1. Data Preparation and Representation

The human compositions are provided to the system as abc notation files. Abc notation [11] is a language for notating music which is easily readable by human and machine, with which many compositions have been written. For the system's human compositions a group of 831 Irish reels [12] were used.

The last sixteen beats of the human compositions, a beat being of crotchet duration, is used as the training phrases. This last part is used so that each training phrase has the same position in the composition. The phrases are then split in to thirty two positions where each position represents a slice of time being half a beat or a quaver in duration. Each position then has a value of either a new sounding note's pitch, a hold of the previous value or a rest. Both held notes and rests are represented as both are present in the training phrases. The pitch of a note is represented by numbering pitches on the chromatic scale from C_4 (middle C) to C_6 (two octaves above) inclusive. C_4 is zero, C_4 sharp is one, D_4 is two and so on. Notes in the human composition that fall outside this range are transposed an octave up or down until they fit. While this does limit representation to a range on the chromatic scale it can make learning and generation simpler. This particular range of just over two octaves was chosen as most human examples fell in this range. An absolute pitch representation was chosen over an intervallic representation as it can represent different keys. Holds and rests are represented by twenty five and twenty six respectively. Rhythm is represented implicitly by which position and thus when a new note is to be sounded, a note is held or sustained, and when to rest. However as each position represents a quaver, only notes which are a multiple of a quaver can be represented. Triplets and semiquavers for example would not fit the scheme, and thus phrases containing such were not used. A phrase and its representation below demonstrates the scheme.
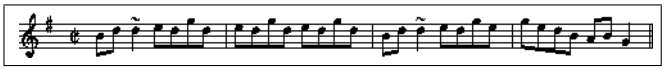


Figure 2. Last four bars from The Primrose Lass by Matt Molloy and Stony Steps.

The phrase of figure 2 is represented as:

11-14-14-25-16-14-19-14-16-14-19-14-16-14-19-14-11-14-14-25-16-14-19-16-19-16-14-11-9-11-7-25

3.2. Neural Networks

The neural network model used is the multilayer perceptron [13]. This neural network is able to find mappings between input data and expected output data. For this project the input data are the prepared music phrases and the output is the phrases class. The multilayer perceptron is also able to generalise. That is given training input/out pairs, the network is able to accurately predict the desired output for an unseen input.

As explained in 3.1, the input data is composed of a set of variables each representing a position of time, the value being either a hold or rest, or the pitch of a new note. Each of these variables are treated as nominal variable. This is as the variable values, pitches, holds and rests, are not assumed to be ordered in a meaningful way. In other words a position having the value of E is not seen as higher or lower than the value of the adjacent F nor the value of a rest or hold. This may seem counter-intuitive, as music theory has ordered scales and intervals which suggest meaningful order between pitches. However treating input variables as nominal resulted in much higher accuracy by the neural networks.

For nominal variables 1-of-N input encoding is suitable [14]. For each variable's possible value or category, there is a separate input neuron which acts as a flag, being either on or off. For each variable there are twenty seven values. Thus for thirty two variables or positions within a phrase, there are then twenty seven times thirty two input neurons, or eight hundred and sixty four input neurons.

The neural network uses only a single layer of neurons. This is as a single layer is all that is needed to capture any

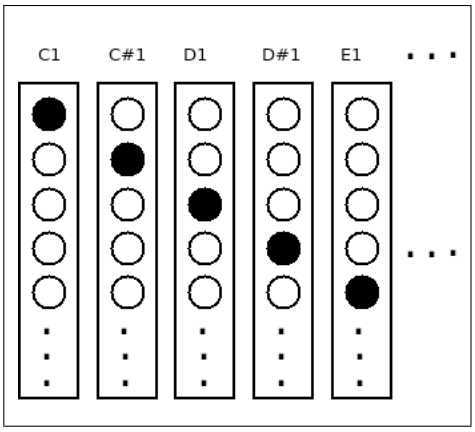


Figure 3. A 1-of-N encoding for single variable. Each circle represents an input neuron, being filled when on.

continuous function to a desired level of accuracy, given enough neurons [15], and in practice additional layers did not improve accuracy. The number of neurons to be used in the hidden layer is to be minimised as less neurons can result in better generalisation. In practice four hidden neurons resulted in accuracy that would not be significantly improved upon by simply adding more hidden neurons. Less hidden neurons performed poorly.

The final output layer has neurons that represent each class, being the examples class and the counter examples class. Training is accomplished by the back-propagation algorithm. The learning and momentum rates are 0.2 and 0.3 respectively, chosen by experimentation for higher accuracy. The learning also decays after each training epoch. This helps avoid the risk of over-fitting the data and thus be unable to generalise well.

3.3. Genetic Algorithm

The genetic algorithm treats phrases as individuals within a population of fifty. From experimentation this level of population was enough diversity to avoid stagnation. To initialise the population phrases are created with random values. An alternative would have been to initialise the phrases as pre-constructed musical phrases [16] or simply from the provided human phrases. However this was not necessary as the genetic algorithm performed well in its search with random initialisation.

The genetic algorithm finds the fittest phrases amongst the population, using the neural networks as a guide. The fittest phrases are those which when inputted to the neural network return the highest confidence for belonging to the set of human phrases. The highest phrases are selected for reproduction and then randomly put into parent pairs. Offspring are created by taking a part from one parent and the rest from the other. The resulting offspring are then mutated by randomly selecting notes, if any, to change to a random value. A dumb, or musically ignorant, mutator was used as opposed to musically aware mutator such as used in GenJam, as it does not require human understanding of a music style nor human preparation of a mutator. The offspring and the parents then form the next generation. This process is repeated for many generations until the highest rating of the population does not increase significantly, an increase of 0.001 over three generations when the algorithm is stopped. This is to avoid spending time when significant improvements are unlikely to occur. Otherwise the algorithm stops after one hundred generations.

The prior described process is used to output the single fittest phrase in the population. While there may be other highly fit phrases they will likely be very similar to the fittest phrase and thus of not much use. For each required phrase the entire process is repeated.

4. SYSTEM RESULTS

The system was run to create the generated phrases. It was found that the processor and memory requirements of the system are quite high, where the time to go through several iterations can be quite lengthy taking approximately a full day using above 3gb of memory. Each iteration add significantly to the time taken. This limited the number of iterations that could be performed, and for the purpose of this paper the system went through six iterations.

For the first part an objective test was conducted to see how well the neural network, or fitness function, would judge varying musical phrases. Secondly, in line with the aim of this project, we need to judge how similar the system-generated phrases are to the human phrases used as training data. We intend to ask human judges to identify which unlabelled phrases are human composed and which are computer generated in order to somewhat objectively measure how well the system performs. Qualitative feedback will also be obtained by asking judges the similarities and differences. However, at this early stage of the project we have conducted only a limited evaluation in which the generated phrases are examined by the authors and the effectiveness of the trained neural networks are tested in some simple experiments. These are described in the following sections.

4.1. Fitness Function Evaluation

Ideally the fitness function should be sensitive to features derived from music theory or readily apparent to most human listeners. These characteristics can include positioning of rests, syncopation, tonality, pitch direction, etc. Those phrase that have the correct characteristics should rate higher than those that don't. Phrase of similar style to the provided human composition should also rate higher than those of different styles.

4.1.1. Test Design

Specific characteristics can be tested for by modifying the human phrases to differ on the given characteristics. These modified phrases should then have lower ratings than the unmodified human phrases themselves. Figure 4 shows the last four bars of The Ewe Reel along with example modifications. In addition the neural networks will also judge sets of 12 bar blues, Swedish polska and Irish jigs to compare different styles, along with random note phrases to provide a base low rating.
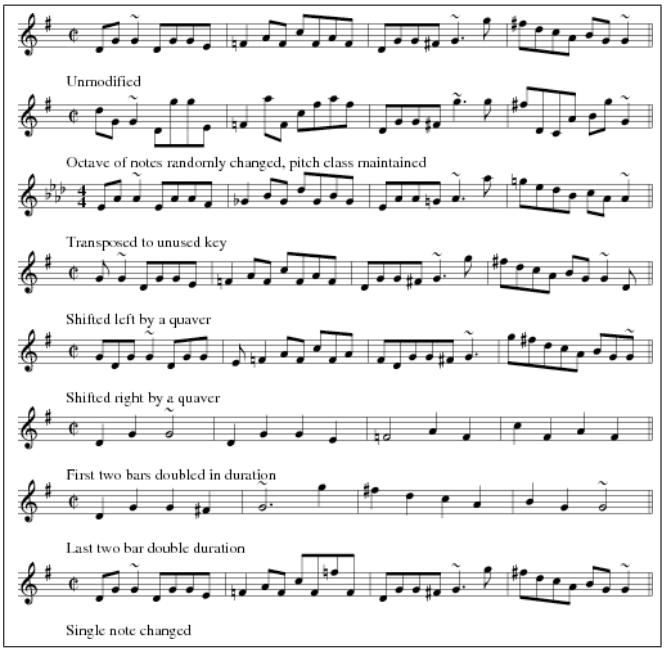


Figure 4. The original phrase and it's modifications

4.1.2. Results

Phrase Set	Average Rating
Unmodified Reels	0.56039
Notes' octave randomly changed	0.25385
Transposed to an unused key	0.10806
Shifted left a quaver	0.49461
Shifted right a quaver	0.50592
First two bars doubled in duration	0.27368
Last two bars doubled in duration	0.26279
Single note changed	0.52469
Twelve bar blues	0.18607
Swedish polska	0.248957
Irish jig	0.39618
Random note phrases	0.09514

Table 1. Results of fitness function evaluation

Through the experiments the unmodified reel phrases received the highest average rating. The ratings show that the system is sensitive to key, octave, note duration and single note changes. However a shift to the left or right resulted in only a small reduction in rating. This suggests that the system is not sensitive to a note's absolute position. It should be noted too that as the phrases are single voiced shifting the phrases does not change the melody position against a backing beat. This means that shifting a melody is less noticable to human ears.

The polska, jig and blues phrases received significantly lower rating than the Irish reel. Particularly interesting is that the Irish jig came the closest to the Irish reels in ratings, perhaps due to the two genres' stylistic similarities. These results suggest that the system is sensitive to musical style.

4.2. Authors' Evaluation

While the authors were not experienced in Irish reels they are experienced musicians, the first being a pianist for many years, the second having a music degree and who performs proffesionally. In their view the system was able produce composition that resemble the given human composed reels, yet the output is still quite recognisably different. Some of the compositions are displayed in figure 5. The following was noted:

- The generated phrases were able to keep within a key. However while the training phrases had many different keys, the generated phrases used a D major or equivalent key. At times the generated phrases would use a note out of key at an unsuitable time.
- Rhythm in both generated and human phrases were similar, with the majority of notes being quavers punctuated by crotchets. Usually the crotchets are on the beat. Rests are uncommon in both.
- The generated phrases at times have more consecutive repeated notes.
- Both human and generated phrases used smaller intervals, with the occasional run. However the generated phrases would at times have a large interval jump caused by a note being an octave too high or too low. Transposing the note up or down an octave is an easy remedy as the out of place notes usually have a suitable pitch class.
- Distinct structure was hard to discern in both human and generated phrases. This is likely as the phrases are four bars long. However this lack of structure was more apparent in the generated phrases.
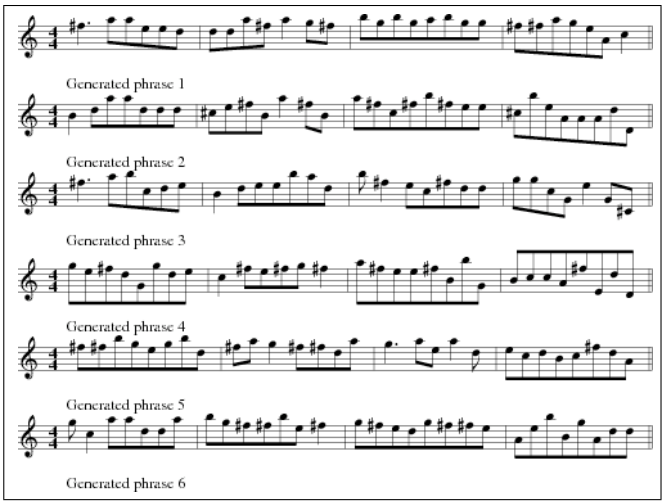


Figure 5. Generated phrases.

5. CONCLUSION

In this paper we have described a system which uses neural networks and evolutionary algorithms to compose music in a style similar to a given set of melodies. This project is in its initial stages but the system shows promise. While the phrases created have musical characteristics and similarities they are clearly distinct from the training phrases, as demonstrated by the evaluation results.

The system generated these phrases without requiring an expert, the author being unfamiliar with Irish reels. The system was also designed with minimal explicit incorporation of musical knowledge. As for flexibility, while only Irish reels were generated for review the system was not designed with a genre in mind. Generated Irish reels whilst resembling the human compositions in some characteristics are distinct.

5.1. Possible Improvements and Future Direction

The representation is currently very limited. Only particular pitches and durations are represented. This shortcoming was most high- lighted by the lack of ornamentation. Elements such as timbre, velocity, articulation, ornamentation, multiple voices and chords are also unrepresented along with pitches outside of the acceptable range. Note durations which aren't multiples of a quaver, such as triplets, are not represented either as they do not perfectly fit the thirty two position representation. The representation could be extended to include and express many of these elements.

The representation is also limited in the generated phrases' length. This length can be extended however this increases the complexity of the task and thus processing time. An alternative would be to treat phrases as elements to be combined into larger compositions.

The system has only been trained on Irish reel. It will be interesting to see it's results given other styles of music. This may require changes to the representation for example for many styles this might require extending the representation to include other elements particularly multiple voices, or adjusting the representation size to fit different meters.

6. REFERENCES

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